

Stabilizer states are theta series

Michael Ontiveros

1 Qudits

An N qudit state over a finite field $\mathbf{F}$ is a vector in the projective Hilbert space

$$\mathbf{H} = \mathbf{C}[\mathbf{F}^N]/\mathbf{C}^*$$

where $\mathbf{C}$ is the field of complex numbers. Assume $\mathbf{F}$ has odd prime order p .

2 Pauli group

The Pauli group is a finite subgroup of the automorphism group of $\mathbf{H}$. It is generated by the linear operators Z_i, X_i defined by

$$Z_i|n\rangle = q^{n_i}|n\rangle \quad X_i|n\rangle = |n+e_i\rangle$$

for $i = 1, \dots, N$ and $n \in \mathbf{F}^N$, where $q = e^{2\pi i/p}$ is a p th root of unity and $e_i = (\delta_{ij})_j$ is the standard i th basis vector in $\mathbf{F}^N$. The generators have order p . They satisfy the commutation relations

$$Z_i Z_j = Z_j Z_i \quad X_i X_j = X_j X_i \quad Z_i X_j = q^{\delta_{ij}} X_j Z_i.$$

3 Stabilizer states

Stabilizer states are N qudit states fixed by N independent Pauli operators.

Theorem. Stabilizer states are theta series

$$\Theta = \sum_{n \in \mathbf{F}^N} |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \quad (1)$$

where $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is a symplectic matrix over $\mathbf{F}$ and u is a vector in $\mathbf{F}^N$.

Proof. We will show that $P_i\Theta = \Theta$ for Pauli operators

$$P_i = q^{u'_i} \prod_{j=1}^N Z_j^{A_{ij}} X_j^{B_{ij}}$$

for $i = 1, \dots, N$ where $u' = u - \text{diag}(AB^t/2)$.

Use the symplectic identities

$$AB^t = BA^t \quad DA^t - CB^t = I$$

to shift the summation variable n .

$$\begin{aligned} P_i\Theta &= \sum_n |(n+e_i)B-uD\rangle q^{\binom{nAB^tn^t/2-uCB^tn^t}{2} + \binom{(n+e_i)B-uD}{2} A^t e_i^t + \binom{u-\text{diag}(AB^t/2)}{2} e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{nAB^tn^t/2 + (e_iAB^t-uCB^t)n^t + (e_iAB^t/2+u-uDA^t)e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{nAB^tn^t/2 + (e_iAB^t-uCB^t)n^t + (e_iAB^t/2-uCB^t)e_i^t} \\ &= \sum_n |(n+e_i)B-uD\rangle q^{(n+e_i)AB^t(n+e_i)^t/2 - uCB^t(n+e_i)^t} \\ &= \Theta \end{aligned}$$

□

4 History

In (1996), Daniel Gottesman introduces stabilizer states. In (2004), he and Scott Aaronson describe an efficient simulation algorithm. In (2006), David Gross proves that stabilizer states are theta series (he calls them Gaussians). In (2021), Craig Gidney modifies the algorithm to simulate deterministic measurements in linear time. Definition 1 is new.

5 Clifford group

The Clifford group is a finite subgroup of the automorphism group of $\mathbf{H}$. It is the automorphism group of the stabilizer states. It contains the Pauli group and three additional generators: the quadratic phase gate Q_i and the sum gate S_{ij} and the Fourier gate F_i defined by

$$Q_i|n\rangle = |n\rangle q^{n_i^2/2} \quad S_{ij}|n\rangle = |n(I+E_{ij})\rangle \quad F_i|n\rangle = \sum_{n'_i} |n(I-E_{ii})+n'_i e_i\rangle q^{n_i n'_i}$$

where E_{ij} is the matrix with one at ij and zero everywhere else.

6 Symplectic representation

Collect the symplectic matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{F}^{(N+N) \times (N+N)}$ and the phase vector $u \in \mathbf{F}^N$ and any other vector $v \in \mathbf{F}^N$ into the affine symplectic matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ u^t & A & B \\ v^t & C & D \end{pmatrix}.$$

The affine symplectic matrices form a group $\widehat{\text{Sp}}$. Definition 1 extends to a map

$$\begin{aligned} f : \widehat{\text{Sp}} &\rightarrow \mathbf{H} \\ M &\mapsto \Theta_M. \end{aligned}$$

Write Th for the image of $\widehat{\text{Sp}}$ in $\mathbf{H}$. The pullback $P_g : \widehat{\text{Sp}} \rightarrow \widehat{\text{Sp}}$ of a Clifford gate g along the map f makes the diagram

$$\begin{array}{ccc} \widehat{\text{Sp}} & \xrightarrow{f} & \text{Th} \\ P_g \downarrow & & \downarrow g \\ \widehat{\text{Sp}} & \xrightarrow{f} & \text{Th} \end{array}$$

commute.

Theorem. Clifford gates pull back to matrices in $\widehat{\text{Sp}}$. They act on the right.

g	Z_i	X_i	Q_i	S_{ij}	F_i
P_g	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ e_i^t & 0 & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ -e_i^t & I & 0 \\ 0 & 0 & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ 0 & E_{ii} & I \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I - E_{ji} & 0 \\ 0 & 0 & I + E_{ij} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & I - E_{ii} & -E_{ii} \\ 0 & E_{ii} & I - E_{ii} \end{pmatrix}$

Proof. Qudit states live in projective space, so it suffices to show that

$$g\Theta_M = K_g \cdot q^{L_g} \Theta_{MP_g}$$

for each Clifford gate g , for constants $K_g \in \mathbf{C}^*$ and $L_g \in \mathbf{F}$.

Z gate

$$\begin{aligned}
Z_i \Theta_M &= \sum_n |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)e_i^t} \\
&= \sum_n |(n - e_i D^t)B - uD\rangle q^{(n - e_i D^t)AB^t (n^t - De_i^t) / 2 - uCB^t (n^t - De_i^t) + e_i B^t (n^t - De_i^t) - uDe_i^t} \\
&= \sum_n |(n - e_i D^t)B - uD\rangle q^{nAB^t n^t / 2 + e_i (I - D^t A)B^t n^t - uCB^t n^t + L} \\
&= q^L \sum_n |(n - e_i D^t)B - uD\rangle q^{nAB^t n^t / 2 - e_i B^t CB^t n^t - uCB^t n^t} \\
&= q^L \sum_n |nB - (u + e_i B^t)D\rangle q^{nAB^t n^t / 2 - (u + e_i B^t)CB^t n^t} \\
&= q^L \Theta_{MP_{Z_i}}
\end{aligned}$$

X gate

$$\begin{aligned}
X_i \Theta_M &= \sum_n |nB - uD + e_i\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\
&= \sum_n |(n - e_i C^t)B - (u - e_i A^t)D\rangle q^{(n - e_i C^t)AB^t (n^t - Ce_i^t) / 2 - (u - e_i A^t)CB^t (n^t - Ce_i^t) + L} \\
&= q^L \sum_n |(nB - (u - e_i A^t)D)\rangle q^{nAB^t n^t / 2 - (u - e_i A^t)CB^t n^t} \\
&= q^L \Theta_{MP_{X_i}}
\end{aligned}$$

Q gate

$$\begin{aligned}
Q_i \Theta_M &= \sum_n |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)E_{ii}(B^t n^t - D^t u^t) / 2} \\
&= \sum_n |nB - uD\rangle q^{n(A + BE_{ii})B^t n^t / 2 - u(C + DE_{ii})B^t n^t + L} \\
&= q^L \Theta_{MP_{Q_i}}
\end{aligned}$$

S gate

$$\begin{aligned}
S_{ij} \Theta_M &= \sum_n |(nB - uD)(I + E_{ij})\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\
&= \sum_n |nB(I + E_{ij}) - uD(I + E_{ij})\rangle q^{n(A(I - E_{ji}))(B(I + E_{ij}))^t n^t / 2 - u(C(I - E_{ji}))(B(I + E_{ij}))^t n^t} \\
&= \Theta_{MP_{S_{ij}}}
\end{aligned}$$

F gate

If the vector $e_i B^t$ is zero, then B has a nontrivial kernel vector $w \in \mathbf{F}^N$. Use it to break up the sum.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n \in \mathbf{F}^N \\ n'_i \in \mathbf{F}}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t + (nB - uD)e_i^t n'_i} \\
&= \sum_{\substack{n \in \mathbf{F}^N / w \\ n'_i \in \mathbf{F}}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_{m \in \mathbf{F}} q^{(n+mw)AB^t (n+mw)^t / 2 - uCB^t (n+mw)^t - uDe_i^t n'_i} \\
&= \sum_{n, n'_i} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_m q^{nAB^t n^t / 2 - uCB^t n^t - uDe_i^t n'_i} \\
&= p \cdot \sum_{n, n'_i} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t - uDe_i^t n'_i}
\end{aligned}$$

Change variables $n'_i \mapsto -e_i A^t (n^t + n'_i w^t) + e_i C^t u^t$.

$$\begin{aligned}
F_i \Theta_M &= p \cdot \sum_{\substack{n \in \mathbf{F}^N \\ n'_i \in \mathbf{F}}} |(n + n'_i w)(B(I - E_{ii}) - AE_{ii}) - u(D(I - E_{ii}) - CE_{ii})\rangle \\
&\quad \times q^{nAB^t n^t / 2 - uCB^t n^t + uDE_{ii}(A^t(n^t + n'_i w^t) - C^t u^t)} \\
&= p \cdot q^L \sum_n |(n(B(I - E_{ii}) - AE_{ii}) - u(D(I - E_{ii}) - CE_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t / 2 - u(CB^t - DE_{ii}A^t)n^t} \\
&= \Theta_{MP_{F_i}}
\end{aligned}$$

If the vector $e_i B^t$ is not zero then there is a vector $w \in \mathbf{F}^N$ with $wB = -e_i$. Use it to split the summation variable $n \mapsto n + mw$. Restrict the new n to the hyperplane $nBe_i^t = e_i Be_i^t$ to avoid over-counting.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle \\
&\quad \times \sum_{m \in \mathbf{F}} q^{(n+mw)AB^t(n+mw)^t/2 - uCB^t(n+mw)^t + ((n+mw)B - uD)e_i^t n'_i} \\
&= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(nB - uD)(I - E_{ii}) + n'_i e_i\rangle q^{nAB^t n^t/2 - uCB^t n^t + n'_i(e_i B - uD)e_i^t} \\
&\quad \times \sum_m q^{-m^2 w A e_i^t/2 - m(n'_i + (nA - uC)e_i^t)}
\end{aligned}$$

Change variables $n'_i \mapsto -n'_i w A e_i^t - (nA - uC)e_i^t$.

$$\begin{aligned}
F_i \Theta_M &= \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t/2 - uCB^t n^t - (n'_i)^2 e_i A^t w^t - e_i B E_{ii} A^t (n^t + n'_i w^t) + u D E_{ii} A^t (n^t + n'_i w^t) + L} \\
&\quad \times \sum_m q^{-m^2 w A e_i^t/2 + m n'_i w A e_i^t}
\end{aligned}$$

The inner sum is a quadratic Gauss sum. It is equal to $K \cdot q^{(n'_i)^2 e_i A^t w^t/2}$ for some constant $K \in \mathbf{C}^*$.

$$\begin{aligned}
F_i \Theta_M &= K \cdot q^L \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{nAB^t n^t/2 - uCB^t n^t - (n'_i)^2 e_i A^t w^t/2 - e_i B E_{ii} A^t (n^t + n'_i w^t) + u D E_{ii} A^t (n^t + n'_i w^t)} \\
&= K \cdot q^L \sum_{\substack{n, n'_i \\ nBe_i^t = e_i Be_i^t}} |(n + n'_i w)(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{(n+n'_i w)(A(I - E_{ii})B^t - B E_i A^t)(n^t + n'_i w^t)/2 - u(C(I - E_{ii})B^t - D E_{ii} A^t)(n^t + n'_i w^t)} \\
&= K \cdot q^L \sum_n |n(B(I - E_{ii}) - A E_{ii}) - u(D(I - E_{ii}) - C E_{ii}))\rangle \\
&\quad \times q^{n(A(I - E_{ii}) + B E_{ii})((I - E_{ii})B^t - E_i A^t)n^t/2 - u(C(I - E_{ii}) + D E_{ii})((I - E_{ii})B^t - E_i A^t)n^t} \\
&= \Theta_{MP_{F_i}}
\end{aligned}$$

□

7 Example

The Clifford circuit

$$Z_1^3 Q_1^4 S_{12} F_1 X_2 |00\rangle$$

generates the twisted EPR pair

$$\Psi = \sum_{n \in \mathbf{F}} |n \ n+1\rangle q^{2n^2+3n}.$$

The symplectic representation is the matrix $P_{X_2} P_{F_1} P_{S_{12}} P_{Q_1^4} P_{Z_1^3} \in \widehat{\text{Sp}}$ defined by the product

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 4 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

which is equal to

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -3 & -4 & 0 & -1 & -1 \\ -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The phase vector and the symplectic blocks are

$$u = (-3 \ -1) \quad A = \begin{pmatrix} -4 & 0 \\ -1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The theta series is

$$\begin{aligned} \Theta &= \sum_{n \in \mathbf{F}^2} |nB - uD\rangle q^{nAB^t n^t / 2 - uCB^t n^t} \\ &= \sum_{n_1, n_2} |(n_1 \ n_2) \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} - (-3 \ -1) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\rangle \\ &\quad \times q^{(n_1 \ n_2) \begin{pmatrix} -4 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} / 2 - (-3 \ -1) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}} \\ &= \sum_{n_1, n_2} |-n_1 \ 1 - n_1\rangle q^{2n_1^2 - 3n_1} \\ &= p \cdot \Psi. \end{aligned}$$

References

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